Carl P. Zimmerman
Briar Creek Tech
Charlotte, North Carolina, USA

---

31 July 2026

To the Editors, *Foundations of Physics*

I submit for your consideration a short research article, *"The MOND acceleration scale is one half of the dark-energy gravitational rate."*

Why this journal. The substance of this paper is conceptual rather than observational, which is why I am offering it to *Foundations of Physics* rather than to a phenomenology journal. The acceleration scale a_0 that organizes galactic rotation curves has been known since 1983 to lie near cH_0, and several closed forms have been proposed for the coincidence. What I add is not a better fit — I am explicit that no improvement in fit is claimed — but a clarification of what the coincidence actually asserts: written correctly, it is a statement about *one rational number*, and the apparent geometric structure that has attached to it dissolves under an elementary reduction. Whether a dimensionless coefficient of order unity relating a galactic dynamical scale to the cosmological constant can be derived, or must be accepted as a brute fact, seems to me a foundational question rather than an astronomical one.

The result.

- $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda} = c^2\sqrt{\Lambda/32\pi} = cH_\Lambda/\sqrt{32\pi/3}$. The three forms are algebraically identical, and the reduction cancels every factor of π and the 3. The consequence I want to draw attention to is negative: $\sqrt{32\pi/3}$ is *only* a unit conversion between H_Λ and $\sqrt{G\rho_\Lambda}$, so any attempt to explain it as a solid angle, a horizon count, or a degree-of-freedom tally is reading significance into arithmetic. I make that point partly because it is an error I made myself.
- On Planck 2018 inputs the expression gives $9.43 \times 10^{-11}\,\mathrm{m\,s^{-2}}$, which is 0.70% from the value obtained by fitting 175 SPARC rotation curves at a stellar mass-to-light ratio $\Upsilon_{[3.6]} = 0.70$ — inside the range expected for Spitzer 3.6 μm populations, so the agreement does not require an implausible fit.
- Section V reports a *failed* attempt to derive the coefficient from ghost-freedom, unitarity and a holographic degree-of-freedom count. I include the failure because the interesting content of the paper is the sharpness of the remaining gap, not a claim to have closed it.
- Section VI gives a falsifiable consequence: if $a_0 \propto \sqrt{\rho_\Lambda}$ then $a_0(z)$ is exactly constant for $w = -1$ and rises to a maximum before *declining* for the DESI-preferred $w_0 > -1$, $w_a < 0$, against the monotonic rise predicted by $a_0 \propto cH(z)$. A measured monotonic rise would exclude the relation.

What I do not claim, stated here because it bounds the paper. The interpolating function I use is *identical* to Eq. (9) of Milgrom (Phys. Lett. A **253**, 273, 1999) — not a variant of it — and that same paper *derives* $a_0 = 2cH_\Lambda$, a factor 11.58 larger. My contribution is therefore a renormalization of his coefficient to match data, not a new derivation, and Sec. II says so in those words. Further, the coefficient here is not observationally distinguishable from Milgrom's later empirical $cH_\Lambda/2\pi$: the two differ by 7.9%, which is 0.018 dex in g_{obs} against a fit scatter of 0.108 dex. Section VII hands over the outstanding tensions unprompted, including that the cluster-scale discrepancy is 13% *worse* for this lower a_0 than for the conventional value. I would rather a referee found these stated than discovered them.

Reproducibility, competing interests, and AI disclosure. Every numerical claim is produced by a public script that performs internal consistency checks and exits with a nonzero status if any fails; the code is archived at `https://doi.org/10.5281/zenodo.21724575`. I am an independent researcher, have no funding to declare, and have no competing interests. Section VIII of the manuscript discloses that portions of the analysis and drafting were carried out with the assistance of a large language model: I directed the work, specified and reviewed every load-bearing calculation, and take full responsibility including for any errors. No artificial-intelligence system satisfies the criteria for authorship and none is listed as an author. Several intermediate claims produced during the work were found to be incorrect and were withdrawn before submission.

Submission history. This manuscript was submitted to *Physical Review D* on 31 July 2026 and was withdrawn by me before review. It is not currently under consideration at any other journal, and it is not part of a joint submission or a sequence of papers. A preprint is deposited at `https://doi.org/10.5281/zenodo.21724575`; it is not a journal publication. The manuscript is not posted to arXiv at the time of submission, and I will inform the editorial office if that changes.

Suggested referees. The value of this manuscript rests largely on whether its treatment of prior work is judged correct, so I would particularly welcome a referee familiar with the acceleration-scale literature:

- Stacy S. McGaugh, Case Western Reserve University
- Benoît Famaey, Observatoire astronomique de Strasbourg
- Federico Lelli, INAF — Osservatorio Astrofisico di Arcetri
- Harry Desmond, Institute of Cosmology and Gravitation, University of Portsmouth

I have no referees to exclude.

I am content for the manuscript to be assessed as a short paper.

With thanks for your time,

Carl P. Zimmerman
Briar Creek Tech
Charlotte, North Carolina